

Rodrigo Grandy, PhD

Principal Scientist | Group Leader

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Summary

Innovative scientific leader with over 19 years of combined experience in academia and biotech, specialising in Cell and Molecular Biology with emphasis on Cell Therapy, RNA/cDNA technologies, and next-generation sequencing.

Certified in AI and Machine Learning for Data Science by the University of Cambridge, with expertise in neural networks, deep learning frameworks, LLMs, and RAG systems.

Currently leveraging this dual expertise as a **Co-Founder and Scientific Lead** for a private biotech venture (Optima Biolabs). I am driving the transition of an ML-driven platform to improve bio-manufacturing of cell-derived products from concept to pre-seed readiness, focusing on technical de-risking, multi-year OpEx modelling, and the articulation of commercial value propositions for the investment ecosystem.

Skilled in establishing and managing R&D teams and projects across cell therapy and nucleic acid sequencing platforms, now enhanced by cutting-edge AI/ML capabilities for advanced data analysis and predictive modelling.

Strong combination of technical expertise spanning wet-lab methodologies and machine learning, with focus on data-driven decision-making, stakeholder management, and leveraging AI tools to unlock insights from complex datasets.

Core Expertise: Strategic R&D & Team Leadership • Entrepreneurial Strategy & OpEx Modelling • iPSC and Hematopoietic Stem Cell Systems • Genomics, Bioinformatics & Sequencing • AI & ML (Neural Networks, Deep Learning, LLMs, RAG) • Data Analysis & Interpretation (NGS, Single-Cell RNA-Seq).

Professional History

Optima Biolabs | Biotech Startup (Pre-incorporation) October 2025 – Present | Oxford, UK
Co-Founder & Scientific Lead (Private Venture)

Project Scope: Leading the formation of a biotech venture leveraging ML and AI to improve bio-manufacturing of cell-derived products. Driving the transition from initial concept to pre-seed readiness through technical strategy, business design, and operational planning.

- Authored comprehensive pitch and due diligence decks; successfully articulated the scientific rationale, market opportunity, and technical execution plan for specialised growth factor disruption.
- Engineered detailed OpEx models, including laboratory setup budgets, specialised equipment procurement, and 24-month financial projections to ensure capital efficiency.
- Orchestrating co-founder recruitment and team formation; evaluating technical synergies and leadership alignment to build a multi-disciplinary founding team.
- Conducted deep-dive competitive landscape analysis to define market entry pathways and differentiation within the cell therapy supply chain.
- Defining the technical and regulatory milestones required to move from proof-of-concept to Series A, including safety compliance and resource scaling (In progress).
- Identifying and addressing critical-path technical and financial risks through iterative reviews with domain experts, KOLs, and potential investors (In progress).

- Cultivating a network of advisors and partners within the Oxford biotech ecosystem to validate the value proposition and identify partnership opportunities.

Studying ML & AI at Cambridge University, PACE, United Kingdom (February 2025 - November 2025)

Over an intensive 9-month journey, I've deepened my expertise across a comprehensive range of data science disciplines:

Unsupervised Learning:

- Anomaly detection and customer segmentation using K-means clustering, PCA, t-SNE, One-Class SVM, and Isolation Forest

Supervised Learning:

- Built predictive models using Random Forest, XGBoost, and Neural Networks
- Applied these techniques to real-world challenges like student dropout prediction
- Model Optimisation & Evaluation:
 - Trained and fine-tuned models for optimal performance
 - Evaluated results using F1-score, recall, accuracy, ROC-AUC, and other key metrics
- Natural Language Processing & Deep Learning:
 - Sentiment analysis and topic modelling
 - Leveraging transformers and LLMs for advanced semantic analysis
- Time Series Forecasting:
 - Mastered advanced methods including SARIMAX, LSTMs, LightGBM, and hybrid approaches
- Final Project: As part of this course I had the privilege of leading a team project in partnership with the Bank of England to solve a real-world regulatory challenge. Our AI-Powered Solution: We developed an intelligent RAG (Retrieval-Augmented Generation) chatbot

Evotec, United Kingdom | Germany (August 2023 - November 2024)

Principal Scientist, Project Lead, Line Manager: iPSC Cell Therapy/Immuno-Oncology

- Managed and supervised a team of scientists and technical staff to ensure the quality of work met company standards in iPSC-derived immunotherapies generation.
- Led scientific strategy and process optimisation for differentiating, expanding, and cryopreserving iPSC-derived HSCs and T cells.
- Contributed to strategic R&D planning: Identified strategies with the VP of immune oncology to control R&D expenses and FTEs within budget.
- Facilitated teamwork among various sites to improve Quality Control processes for iPSC maintenance, expansion, and MCB production.
- Coordinated inter-site actions to optimise the differentiation of hematopoietic stem cells from iPSCs.
- Spearheaded the creation of an mRNA-seq single cell atlas for internal use, focusing on immune cell differentiation during human embryonic development.
- Worked closely with the Business Development Director and SVP of Cell Therapy to showcase the company's iPSC-derived T cells program and cell therapy manufacturing capabilities to potential R&D and manufacturing corporate clients.
- Served as interim Line Manager for scientists leading iPSC-derived macrophage development, overseeing activities, setting quarterly objectives, and providing mentorship to enhance team performance.
- Partnered with senior leaders on key projects, including iPSC line generation in a GMP environment and establishing NGS platforms for assessing iPSC line genetic stability.

- Developed a comprehensive strategy for biomarker assay generation using flow cytometry to characterise cell types during T cell differentiation and intermediate stages.
- Supervised the characterisation of iPSC-derived T cell phenotypes and functionality, led by the immunology team in Toulouse, France.

Adaptimmune | Abingdon, UK (February 2022 - August 2023)

Group Leader/Line Manager Stem Cell Research

- Developed a quality control package in compliance with international standards to assess the quality of in-house derived iPSC cell lines.
- Formed and led a team of senior scientists and scientists to manage the quality control program for our engineered iPSC cells and derivatives.
- Conducted performance reviews and addressed any misalignment of team members with group objectives.
- Reported progress to stakeholders. Collaborated with project leads and other departments to ensure timely delivery of results.
- Directed the research program, improving the efficiency and consistency of iPSC cell cultures, genetically engineered cell lines, and HSC differentiation systems to meet company objectives.
- Created and implemented a training program to standardise iPSC cell culture techniques and ensure adherence to international guidelines for scientists.
- Coordinated the quality control, banking, and differentiation processes for hiPSC lines and HSC derivatives, facilitating the generation of allogenic T cells.
- Served as a subject matter expert on stem cell-related matters, providing strategic guidance and advice across organisation.
- Defined and managed the research program budget, ensuring financial resources were aligned with project goals.
- Engaged in constructing and improving the platform to support the development of engineered iPSC-derived T cell therapies.

Cytiva, Cardiff, UK (October 2019 - February 2022)

Senior Scientist Research and Development: Genomics & Cellular Research

- Generated and analysed NGS data to evaluate performance and quality of various Cytiva products and prototypes.
- Operated and maintained Next Generation Sequencer. Collaborated with the head of R&D to implement Single Cell analysis laboratory capabilities.
- Developed a method for detecting COVID-19 viral RNA through "Direct RT-PCR," optimising the formulation for reliable and consistent detection in human nasopharyngeal swab samples.
- Provided advisory support to R&D and G&D leads, helping them understand NGS and Single Cell Technology data, applications, and requirements, while collaborating with leadership to establish a new Single Cell Technology portfolio.
- Partnered with R&D and G&D leadership to establish new Single Cell Technology lab, expanding company's capabilities.
- Collaborated with the Lead Chemist to develop streptavidin-conjugated magnetic beads for use in hybridisation capture assays, including Exome sequencing.
- Supervised two junior scientists on a nucleic acid isolation project and a scientist on the Single Cell Technology portfolio.

- Led the team for Single Cell, NGS, and New Isolation Kit projects, overseeing junior scientists and supporting M4 activities for the VIA Omics Bundle.

Education, Training, and Certifications

- AI and Machine Learning for Data Science Certification
University of Cambridge (UK) | 2025
- Postdoctoral Researcher
Stem Cell Institute, Cambridge University (UK) | 2015-2019
- Postdoctoral Researcher
UMass Medical School (USA) | 2011-2015
- PhD in Cell and Molecular Biology
University of Concepcion (Chile) | 2011
- Bachelor's Degree in Biochemistry
University of Concepcion (Chile) | 2004

Publications & Conferences

Full list of publications available on: <https://www.ncbi.nlm.nih.gov/myncbi/1xWYmtc7fmqI9z/bibliography/public/>